

WEI-WEI WANG

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🌐 RESEARCH INTERESTS

- **Algebraic Coding:** Quantum MDS codes, Error-correcting codes in information security.
- **Cryptography:** Secure multi-party computation, Homomorphic encryption and so much more.
- **Service Computing:** Recommendation system, QoS prediction, etc.
- **TCM AI:** Diagnosis and treatment assistance (practitioner-oriented AI assistant).

🎓 EDUCATION

Xidian University (XDU), Xi'an, Shaanxi, China Sept. 2021 – Present

PhD student in Cryptography, expected Jun. 2025

Liaoning University (LNU), Shenyang, Liaoning, China Sept. 2018 – Jun. 2021

MSc in Fundamental Mathematics

Baicheng Normal University (BCNU), Baicheng, Jilin, China Sept. 2014 – Jun. 2018

BSc in Mathematics and Applied Mathematics

📖 PUBLICATIONS

- J5 **Weiwei Wang**, Wenping Ma, Kun Yan. Trust-aware privacy-preserving QoS prediction with graph neural collaborative filtering for internet of things services. *Complex & Intelligent Systems*, 2025, 11(4), 191. [\[Paper\]](#)
- J4 **Weiwei Wang**, Wenping Ma, Kun Yan. FSPPCFs: A Privacy-Preserving Collaborative Filtering Recommendation Scheme Based on Fuzzy C-means and Shapley Value. *Complex & Intelligent Systems*, 2025, 11(1), 107. [\[Paper\]](#)
- C1 **Weiwei Wang**, Jiantao Li. Some new classes of quantum convolutional and asymmetric MDS codes. In *Proceedings of the International Conference on Modeling, Natural Language Processing and Machine Learning (CMNM '24)*, 2024, 160–166. [\[Paper\]](#)
- J3 **Weiwei Wang**, Jiantao Li. Two classes of entanglement-assisted quantum MDS codes from generalized Reed–Solomon codes. *Quantum Information Processing*, 2022, 21(7), 245. [\[Paper\]](#)
- J2 Kun Yan, Wenping Ma, Qi Yang, Shaohui Sun, **Weiwei Wang**. Info-Chain: Reputation-Based Blockchain for Secure Information Sharing in 6G Intelligent Transportation Systems. *IEEE Internet of Things Journal*, 2024, 11(5), 9198–9212. [\[Paper\]](#)
- J1 Qizheng Wang, Wenping Ma, **Weiwei Wang**. B-LNN: Inference-time linear model for secure neural network inference. *Information Sciences*, 2023, 638: 118966. [\[Paper\]](#)

👥 PROJECT EXPERIENCE

MPC-HMAC Project Jun. 2021 – Aug. 2022

This is the first project I worked on with Safeheron Lab.

Brief introduction: For the HMAC protocol, based on the MPC (Secure Multi-Party Computation) theory, we conduct research and development on the MPC-HMAC scheme. Mainly based on the HMAC signature algorithm, we adopt the two-party MPC theory, combine the existing hashing algorithms, Garbled Circuit, Oblivious transfer, elliptic curve cryptography, and other related research results, design and develop the MPC-HMAC two-party signature scheme and the MPC-HMAC two-party private key sharding generation and rotation scheme, and investigate the multi-party MPC-HMAC signature protocol.

Research Topics:

- MPC-HMAC two-party private key shard generation scheme
- MPC-HMAC two-party private key shard rotation scheme

- MPC-HMAC two-party signature scheme

Homomorphic Computation of Non-polynomial Functions

Aug. 2022 – Oct. 2022

This is my project for the 2022 National “Financial Cryptography Cup”.

Brief introduction: The specific research topic is how to perform homomorphic calculations on non-polynomial functions. For example, the inverse function and the ReLU function. For the inverse function, we rely on the Taylor expansion of the exponential function e^{-tx^2} , and obtain the approximate polynomial of x^{-1} through transformation, and implement it based on the HELib algorithm library. The ReLU function relies on the Taylor expansion of the Logistic function, and obtains the approximate polynomial of the ReLU function through transformation, and implements it based on the HEAAN algorithm library.

⚙️ SKILLS

- Programming Languages: Python, C++, $\text{\LaTeX}$, MATLAB, Linux
- Platform: PyTorch, Tensorflow, NumPy, Anaconda

♥️ SELECTED HONORS AND AWARDS

Nov. 2024 Second Prize for Postgraduate Academic Scholarship.

Feb. 2023 Outstanding Individual Award on “Financial Cryptography Cup” National Cryptography Application and Technology Innovation Competition.

Feb. 2023 Team Third Award on “Financial Cryptography Cup” National Cryptography Application and Technology Innovation Competition.

Oct. 2023 Second Prize for Postgraduate Academic Scholarship.

Oct. 2022 Second Prize for Postgraduate Academic Scholarship.

Oct. 2021 Second Prize for Postgraduate Academic Scholarship.

May. 2021 Outstanding Graduate Student and Second Postgraduate Academic Scholarship of LNU.